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1 % In the preamble \usepackage{mathtools}
2 \def\B{\mathcal B}
3 \fboxsep=0pt
4 % Placeholder to mimic preceding expressions
5 \def\precexp{\fbox{\phantom{aaa}}}
6
7 % Standard approach with no concern for spacing
8 \begin{alignat}{3}
9 &\precexp\int_{\B(0,1)} f(x)\,, dx &&\quad\\
10 &\precexp\int_{-\infty}^{+\infty} g(x)\,, dx\\
11 &\precexp\sum_{1\leq i + j + k\leq N} N \\
12 &\quad (x_i + x_j) x_k &&\quad\\
13 &\precexp\sum_{i=1}^{\phi(g(i))=N} i! \\
14 \end{alignat}
15
16 % In the next page you'll see the action of the
17 % macro \limits, which puts your (sub/super)script
18 % box under the operator. The macros \mathllap,
19 % \mathclap and \mathrlap from mathtools create
20 % left, centered or right overlapping boxes with
21 % their content along with the adjacent operator.
22 % The effect is to ignore the width of subscripts
23 % and superscripts without having to mess with
24 % the space manually.

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25
26 % Here you can try also \mathclap and see the
27 % difference
28 \begin{alignat}{3}
29 &\&\precexp\int\limits_{\mathrlap{\B(0,1)}}
30 &f(x)\,,\, dx &&\&\{\qquad\}
31 \&\precexp\int\limits_{\mathrlap{-\infty}}
32 &\limits_{\mathrlap{+\infty}} g(x)\,,\, dx
33 &\tag{1.1}
34 \end{alignat}
35
36 % These macros work great when the heights of the
37 % boxes surrounding the operators are small compared
38 % to the ones of the operators; otherwise they
39 % might cause more harm than good. In that case,
40 % consider assigning a symbol to your
41 % sub/superscript material and use that instead.
42
43 \begin{alignat}{3}
44 &\&\precexp\sum_{\mathclap{1\leq i + j + k\leq N}}^N
45 &(x_i + x_j) x_k &&\&\{\qquad\}
46 &\&\precexp\sum_{i=1}^{\mathclap{\phi(g(i))=N}} i!
47 &\tag{2.1}
48 \end{alignat}

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